Walmart Midterm Project

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The Mess

60 ways to write 12 stores

The same store showed up under many different IDs, so sales from one store looked like they came from several.

- E.g., FL-412, fl_412, FL412, FL-412...
- 4,345 rows needed a fixed store ID
- Without it, sales couldn't join to store locations

5 timestamp formats

Dates and times were written in five different ways, and some rows had no time at all.

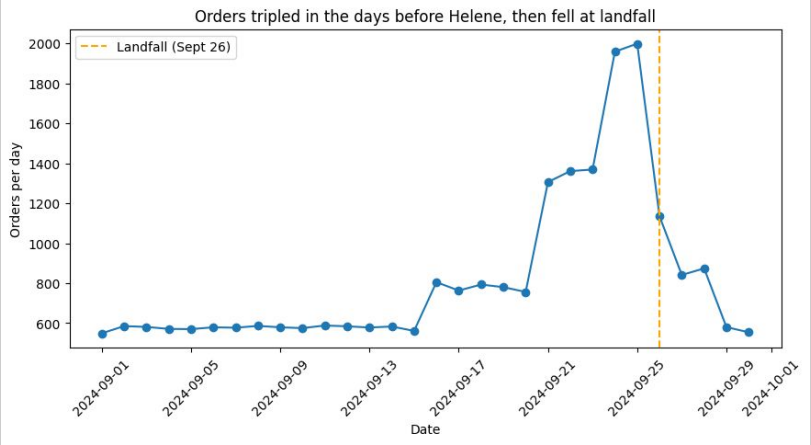
- E.g., 2024-09-25 08:15 vs 09/25/2024 20:08...
- A basic date parse read only about 10% of rows
- 886 rows had no timestamp

4 SKUs for one Pop-Tart

Strawberry Pop-Tarts was recorded under four product codes, which hid how much it really sold.

- E.g., PT-12, SKU#459812, POPTART-STRAW, PT_STRAWBERRY
- The split up shows the best Pop-Tarts SKU sold only about 103 a day, rank #3 - underestimated!
- After combination Pop-Tarts sold about 377 a day, rank #1!

The Pipeline



Clean

- 60 store IDs → 12 stores, 37 categories → 10
- 45 SKUs → 20 real products, all 5 date formats parsed
- 22 cleaning steps logged, each with a row count

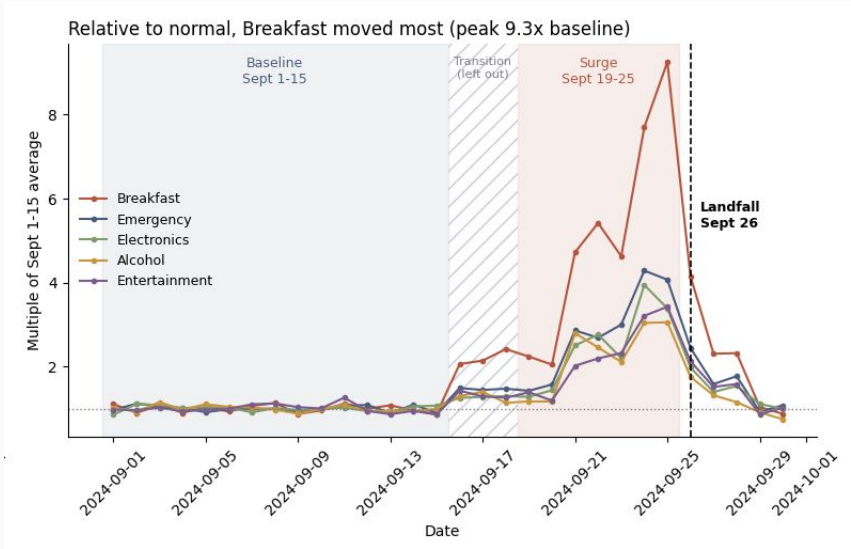
Join Weather

- Open-Meteo API weather for all 12 store locations
- Joined on date + store location
- 3,600 rows before the join, 3,600 after - none lost

Flag & Keep

- Only 30 exact duplicates removed (25,562 → 25,532)
- 884 rows with no timestamp kept, left out of time charts
- 130 negative quantities kept as returns, left out of demand

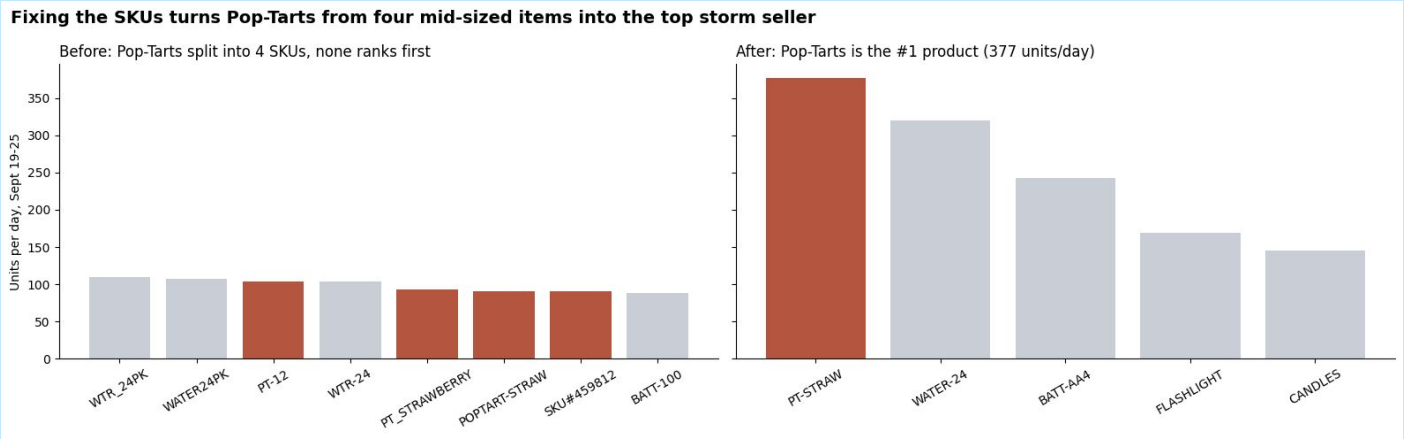
Q1: Demand Surge



Breakfast Surged 414% Before Landfall

- We compared normal days (Sept 1–15) with the 7 days before landfall (Sept 19–25). We left out Sept 16–18 because sales were already starting to rise.
- Breakfast had the biggest jump, from 73 to 377 units a day (+414%). At its peak on Sept 25, it sold 9.3 times its normal amount.
- Emergency supplies (+184%) and Electronics (+150%) came next. Emergency was the biggest category overall, selling about 650 more units a day.
- Every top category peaked on Sept 24–25 and dropped once the storm hit.

Q2: The Pop-Tarts Effect



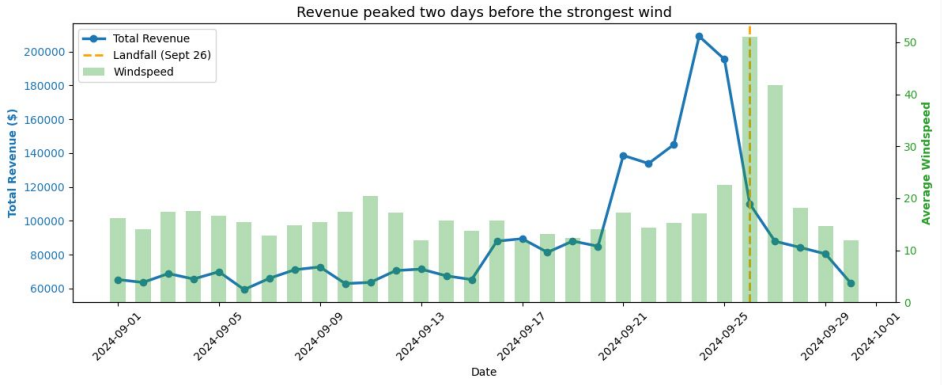
Before: Split Into 4 SKUs

- The best Pop-Tarts code sold about 103 a day, ranked #3 behind water
- Looks like an ordinary item, so it would be under-ordered

After: Combined Into One Product

- 377 a day in the storm week, the #1 storm product
- Ahead of bottled water (320 a day)

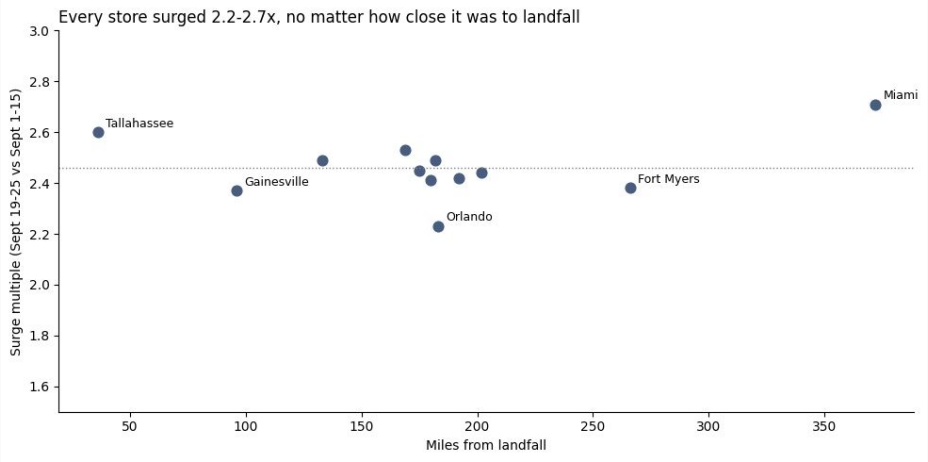
Q3: Weather Correlation



People Bought About 2 Days Before the Worst Weather

- We pulled real daily weather from the Open-Meteo API for all 12 store locations and joined it to sales by date and store.
- Sales peaked on Sept 24–25, but the strongest wind and heaviest rain both came on Sept 26, the day of landfall.
- Sales and wind barely line up on the same day, but match closely when we compare sales with the wind two days later.
- This means shoppers reacted to the **forecast**, not the weather itself. It's based on 30 days and one storm, so it shows a pattern, not proof.

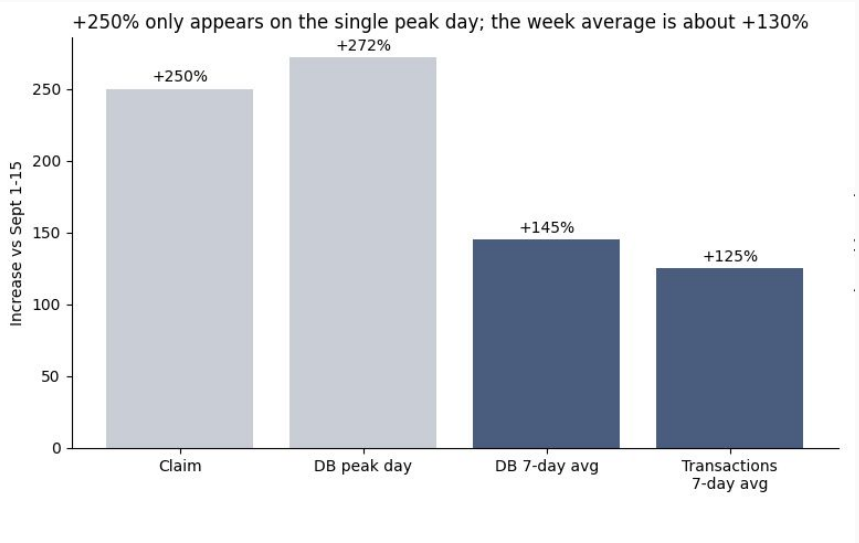
Q4: Store-level Geographic Patterns



Every Store Surged 2.2–2.7x, Near or Far from the Storm

- All 12 stores sold between 2.2 and 2.7 times their normal amount, and all of them surged in the same week.
- Miami, the farthest store from landfall (372 miles), had the biggest surge at 2.71x. Orlando had the smallest at 2.23x.
- Distance from landfall barely made a difference (correlation of 0.23), and with only 12 stores, that could easily be noise.
- Shoppers across Florida reacted to the statewide forecast, not to how close they were to the storm.

Q5: Signal vs. Artifact



The Surge Is Real, but +250% Overstates It

- The +250% number only shows up if you compare the single busiest day (Sept 24) to a normal day.
- Over the full storm week, the increase was about +130% (+125% in the transactions, +145% in the database), mostly Monopoly (+160%).
- The surge itself is real: the sales database and the transactions both show it, and only 10 of 360 database rows had errors.
- Board game stock dropped from about **134 to 33 units** per store right when sales went up, which only happens with real buying.

Recommendation

Restock a Week Early

Sales started rising about 10 days before the storm and peaked on Sept 24–25, just 1–2 days before landfall. Stock that arrives on landfall day is mostly too late, so start restocking a week ahead.

Restock Every Store

All 12 stores surged 2.2–2.7x in the same week, no matter how close they were to the storm. Miami, the farthest store, surged the most, so send extra stock statewide, not just to stores on the storm's path.

Stock Pop-Tarts First

Breakfast rose 414% and Emergency 184%. Pop-Tarts was the #1 storm product once its 4 SKUs were combined, so track it as one item. For Board games, stock more, but plan for about +130%, not +250%.

Split Supply: JAX + Miami

Jacksonville is closest to 9 of the 12 stores and would carry about 74% of the extra demand. Moving Tampa, St. Pete, Lakeland and Kissimmee to the Miami center balances the load, with Atlanta as backup.

What Broke

– The Weather API Timed Out:

– While rerunning the notebook, the Open-Meteo API timed out. Our backup kicked in, but it only had 30 rows of Orlando weather instead of 3,600 rows for all 12 stores, so every chart after it broke.

– What We Learned:

– A backup is only useful if it matches the real data. Next time, we'd save the full store-level weather to a file once and reuse it, and give the API more time before retrying.

Thank you!